AXOLABS PARTIAL METHOD VALIDATION REPORT FOR THE ANALYSIS OF 2439-EXNA IN DOG CSF	
Axolabs Project Number	AN404.0-24
Project Title	Partial validation of an AEX-HPLC-based method for the analysis of 2439-exNA in dog CSF
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Start Date of Bioanalytical Test Phase	20-Sep-2024
End Date of Bioanalytical Test Phase	20-Sep-2024

SIGNATURE PAGE

Name and Function	Date and Signature
Nicole Rottmann **STUDY DIRECTOR**	17-Dec-2024 [redacted]
Name and Function	Date and Signature
Alissa Coffey **SPONSOR'S REPRESENTATIVE**	18-Dec-2024 [redacted]

COMPLIANCE STATEMENT

This *Partial validation of an AEX-HPLC-based method for the analysis of 2439-exNA in dog CSF* was conducted in accordance with the procedures described herein.

The method validation was conducted in compliance with *ICH guideline M10 on bioanalytical method validation and study sample analysis* (EMA/CHMP/ICH/172948/2019) and following the rules in the *OECD Principles of Good Laboratory Practice* (Revision from 1997).

There were no deviations from the Method Validation Plan and respective SOPs. All work was done according to the Method Validation Plan and respective SOPs.

The Method Validation Report represents an accurate and complete record of the results obtained during the method validation.

No deviations from the above regulations affected the overall integrity of this method validation.

Study Director:	Nicole Rottmann	17-Dec-2024 [redacted]
	Name	Date and Signature

QUALITY ASSURANCE STATEMENT

This Method Validation Report titled *Partial validation of an AEX-HPLC-based method for the analysis of 2439-exNA in dog CSF* has been audited.

The reported results accurately reflect the raw data generated during this method validation.

As the bioanalytical testing is of short duration, routine and repetitive, Axolabs GmbH applies a process-based inspection program. The process-based inspection relevant to the activities performed during this method validation is listed below.

Table 1: Quality Assurance Inspections

Type(s) of Inspection(s)	Inspected Activities	Date(s) of the Inspection(s)	Date(s) of Reporting and Recipient(s)
Process-based (Report No.: II-24-003)	• Incoming Goods Check, storage, and labeling of study samples • Incoming Goods Check, storage, and labeling of reference standard, probe, blank matrix • Labeling and preparation of reagents/ solutions • Labelling and preparation of reference standard stock solution • Preparation of calibration standards and QC samples • Sample preparation • Lysate preparation and hybridization • HPLC setup and analysis • HPLC data processing and data evaluation • Disposal of study samples	03, 05, 11 & 18-Sep-2024	TSM: 11-Oct-2024 SD: 15-Oct-2024
Audit of Report and Data	Draft Report Version from 03-Dec-2024	06-Dec-2024	SD: 06-Dec-2024 TSM: 17-Dec-2024

TSM: Test Site Management. SD: Study Director.

In addition to the listed process-based inspections, facility-based inspections take place yearly.

Quality Assurance Axolabs:	Cornelia Schumann	17. Dec. 2024 [redacted]
	Name	Date and Signature

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LIST OF ABBREVIATIONS

AB0X	Analysis Batch, AN404.0-24_01_AB0X
ACN	Acetonitrile
AEX-HPLC	Anion-Exchange High-Performance Liquid Chromatography
BQL	Below Quantification Limit (Below the LLOQ)
CHMP	Committee for Human Medicinal Products
CoA	Certificate of Analysis
Conc.	Concentration
CSF	Cerebrospinal Fluid
CV	Coefficient of Variation
DNA	Deoxyribonucleic Acid
EMA	European Medicines Agency
EU	European Union
FTS	Freeze-Thaw Stability
GCP	Good Clinical Practice
GLP	Good Laboratory Practice
ICH	International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use
LTS	Long-Term Stability
LLOQ	Lower Limit of Quantification
MDR	Method Development Report
n.a.	Not Applicable
OECD	Organization for Economic Co-operation and Development
PNA	Peptide Nucleic Acid
PrP	Prion Protein
PSS	Processed Sample Stability
QA	Quality Assurance
QC	Quality Control
SD	Standard Deviation
SDS	Sodium Dodecyl Sulfate
siRNA	small interfering Ribonucleic Acid
SOP	Standard Operating Procedure
SST	System Suitability Test(ing)
STM	Standard Test Method
STS	Short-Term Stability
TCEP	Tris(2-carboxyethyl)phosphine
ULOQ	Upper Limit of Quantification
UV	Ultraviolet

1 SUMMARY

This report summarizes the results of the bioanalytical partial method validation for the analysis of 2439-exNA in dog CSF (Axolabs Project: AN404.0-24).

The assay is based on the specific hybridization of the antisense strands of 2439-exNA, a dimer composed of two identical siRNAs, with a complementary peptide nucleic acid (PNA) strand in a 1:1 stoichiometry relative to each antisense strand. The PNA is labeled with an Atto425 fluorescence dye and yields a specific signal in the subsequent analysis by Peptide Nucleic Acid Anion-Exchange High-Performance Liquid Chromatography (PNA-AEX-HPLC) coupled to a fluorescence detector.

The method was developed without the need for any analyte extraction from dog CSF. Therefore, recovery of 100 % is maintained over the process (no internal standard is needed).

The bioanalytical method was previously developed at Axolabs GmbH and is described in the Axolabs Method Development Report (MDR) *MDR_2439exNA_CSF* (edition 01, effective date: 21-Aug-2024) that was used as the basis for this method validation (Supplemental File 1). During the method development phase, the suitability of the method for dog CSF was shown.

In addition to the partial method validation, a full method validation for the analysis of 2439-exNA in rat CSF was conducted at Axolabs GmbH (Axolabs Project: AN405.0-24). In the partial and full method validations, artificial CSF was used as surrogate matrix for all validation tests except for selectivity and the matrix effect that will be evaluated using the respective blank matrices (dog or rat CSF). Hence, only selectivity and the matrix effect were assessed in this partial method validation, since all other validation tests would be redundant to those conducted in the full method validation.

The method validation was carried out with the same conditions and parameters described in the MDR, except for the preparation of the reference standard. In the method development a lyophilized reference standard was used, which was weighed and dissolved. In the validation, an already prepared reference standard solution was used, which was provided.

During the method development, two special requirements for this method were identified that were very important for method validation.

The first requirement is to use artificial CSF as surrogate matrix. Beagle dog CSF was not used as blank matrix in the method development, because it is a rare matrix and difficult to purchase. This procedure is in accordance with the ICH M10 guideline and the 3Rs Principle (Replacement, Reduction and Refinement) intended to reduce animal experiments to an absolute minimum. The composition of the artificial CSF is chosen to be as similar as possible to the biological CSF. In terms of animal welfare, artificial CSF is therefore the best alternative to replace biological matrix. In addition, the suitability of the artificial CSF was demonstrated during this method validation as part of selectivity and matrix effect testing – those tests were performed with beagle dog CSF, while the calibration samples were prepared with the surrogate matrix. For all other validation tests, artificial CSF was used.

The second requirement is to prevent non-specific binding effects of the analyte 2439-exNA in CSF by adding a solution of Tween 20 and TCEP. Nonspecific binding is a problem in bioanalytical assays where a compound can adsorb to the container surface, resulting in a reduced recovery of the analyte during analysis. The adsorption happens frequently in CSF because this matrix lacks proteins and lipids that can bind to the analytes or solubilize lipophilic analytes.
To avoid possible adhesion effects in study samples, the planned procedure for sample processing is to add 5 % (v/v) of an 8 % Tween 20 / 50 mM TCEP solution directly to the study samples after thawing and before any pipetting step. This procedure was reflected in the stability testing of the method validation performed under Axolabs project No. AN405.0-24. As this 8 % Tween 20 / 50 mM TCEP solution was also added to all calibration standard and QC samples in the same ratio as in study or LTS samples, no formal dilution of the study/LTS samples had to be calculated.

The final conditions of the validated method are described in the Standard Test Method (STM) *STM_Broad_2439-exNA_DogCSF* (Revision 01, effective date: 04-Oct-2024; Supplemental File 2).

Additional bioanalytical method validations for 2439-exNA conducted at Axolabs are listed in Table 2.

Table 2: Summary Information of additional Method Validations for 2439-exNA

Axolabs Project	Type of Assay	Details of Validation	Date of Final Report
AN402.0-24	PNA-HPLC-Assay with Fluorescence Detection	other matrix	04-Dec-2024
AN403.0-24	PNA-HPLC-Assay with Fluorescence Detection	other species and matrix	03-Dec-2024

The work was done according to the Method Validation Plan signed on 13-Sep-2024 by the Sponsor's Representative.

The results of the method validation are summarized in Table 3.

Table 3: Overall Summary Table

Items	Results	Section/Table	Comments
Analyte	2439-exNA	Section 3	n.a.
Reference Standard	2439-exNA	Section 4	n.a.
Methodology	PNA-AEX-HPLC-Assay with Fluorescence Detection	Section 8.1	n.a.
Biological Matrix	Dog CSF	Section 5 Table 5	n.a.
Calibration Curve Range	1.0 ng/mL – 1,000.0 ng/mL	Section 9.2 Table 11	Test was performed using artificial CSF
Selectivity	5 times higher response (peak area) in the spiked QC-LLOQ samples compared to the signal in the blank samples	Section 9.3 Table 13	Test was performed using 6 different Beagle Dog CSF lots
Matrix Effect	No matrix effect observed for 6 individually tested matrix lots	Section 9.4 Table 14	Test was performed using 6 different Beagle Dog CSF lots
Carry-Over	<20 % of the response (peak area) of the LLOQ calibration standard.	Section 9.5 Table 15	Test was performed using artificial CSF
Within-Run Accuracy	QC-LLOQ (1.0 ng/mL): 111.7 % QC-Low (3.0 ng/mL): 102.2 % QC-Mid (450.0 ng/mL): 99.7 % QC-High (900.0 ng/mL): 97.1 %	Section 9.7.1 Table 16	Test was performed using artificial CSF; Between-Run Accuracy was performed under Axolabs Project No. AN405.0-24
Within-Run Precision	QC-LLOQ (1.0 ng/mL): 6.7 % QC-Low (3.0 ng/mL): 2.7 % QC-Mid (450.0 ng/mL): 3.0 % QC-High (900.0 ng/mL): 5.9 %	Section 9.8.1 Table 16	Test was performed using artificial CSF; Between-Run Precision was performed under Axolabs Project No. AN405.0-24
Dilution Integrity	Test was performed under Axolabs Project No. AN405.0-24		
Long-Term Stability	Test was performed under Axolabs Project No. AN405.0-24		
Freeze-Thaw and Short-Term Stability	Test was performed under Axolabs Project No. AN405.0-24		
Bench-Top Stability (Robustness)	Test was performed under Axolabs Project No. AN405.0-24		
Processed Sample Stability	Test was performed under Axolabs Project No. AN405.0-24		
Short-Term Stability of Stock and Working Solution	Test was performed under Axolabs Project No. AN405.0-24		
Long-Term Stability of Stock Solution	Test was not performed as data were provided by the Sponsor.		

2 COMPLIANCE

The method validation was conducted in compliance with the following guidelines, regulations, or guidance documents:

- ICH guideline M10 on bioanalytical method validation and study sample analysis (EMA/CHMP/ICH/172948/2019); European Medicines Agency; 21 January 2023

Additionally, this method validation project was conducted following the rules as depicted in the *OECD Principles of Good Laboratory Practice* (Revision from 1997).

3 ANALYTE

The analyte 2439-exNA is a prion protein (PrP)-lowering divalent siRNA (Figure 1). The sense strands of the two identical siRNAs are linked to each other via a tetraethylene glycol linker.

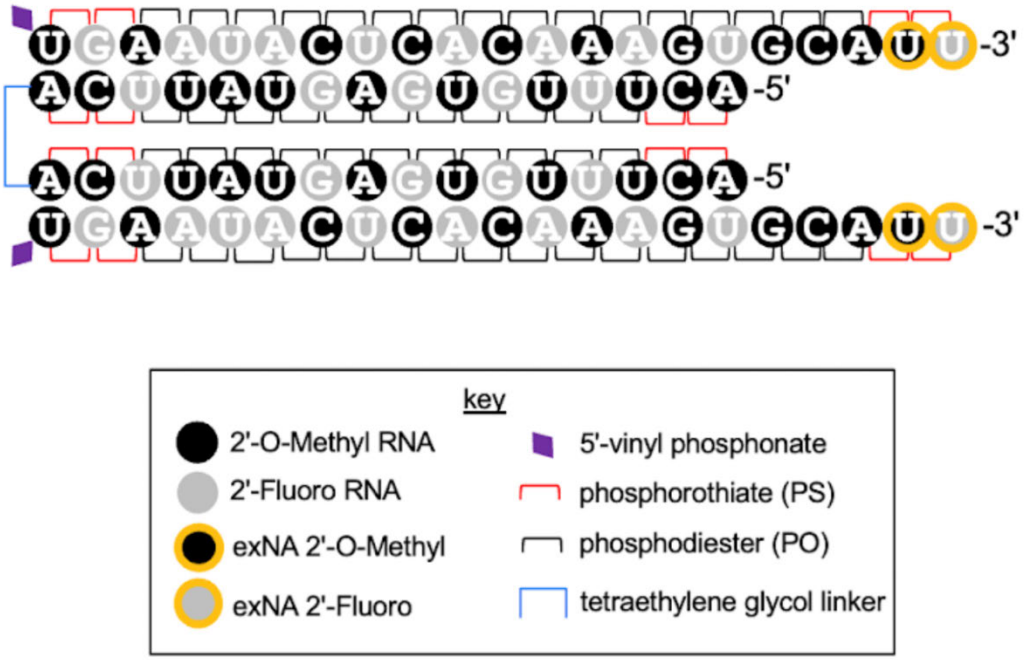


Figure 1: Structure of the PrP-lowering divalent siRNA 2439-exNA

4 Reference Standard

Identification:	PrP-lowering divalent siRNA 2439-exNA (shortened to 2439-exNA in this document)
Structure:	Refer to Section 3
Physical Description:	Slightly yellow solution
Purity:	96.6 % (determined by non-denaturing HPLC)
Total Oligo Concentration:	9.9 mg/mL (determined by UV spectrophotometry)
Full-Length Concentration:	9.5634 mg/mL (corrected for purity stated on the CoA)
Supplier:	Argonaut Manufacturing Services (Carlsbad, CA, USA)
Lot No.:	240621005
Manufacturing Date:	24-Jun-2024
Retest Date:	24-Jun-2026

With the exception of a document-based inspection and control of the corresponding Certificate of Analysis (CoA), no further testing on identity and purity of the reference standard was conducted upon receipt.

The reference standard stock solution was provided by Argonaut Manufacturing Services and was diluted with a solution of 1 % Tween 20 and 1 mM TCEP in artificial CSF to prepare the working solutions for the calibration standards and the Quality Control (QC) samples. Separate aliquots were used for the calibration standards and the Quality Control (QC) samples. For this purpose, the reference standard was previously aliquoted into 100 µL aliquots (Eppendorf Protein LoBind tubes). After aliquoting, the reference standard was renamed to 2439-exNA_240621005_Stock1_04-Sep-2024 in accordance with internal nomenclature. The concentrations of calibration standards and QC samples refer to the full-length concentration of 2439-exNA.

5 Blank Matrices

Blank matrices used in this method validation are listed in Table 4.

Table 4: Blank Matrix for method validation

Blank Matrix	Sex	Receipt Date	Supplier	Lot No.	Expiry Date	Analysis Batch
Artificial CSF	n.a.	12-Aug-2024	VWR International, Darmstadt, Germany	240719	19-Jul-2025	AB01

Beagle dog CSF samples of the non-clinical study with the study protocol no. A-703-02 (Axolabs Project No. AN409.0-24) were used for the selectivity and matrix effect test. Since the control samples were used as a blank matrix for the selectivity and matrix effect test, there is no proven shelf-life on this blank matrix. The Beagle dog CSF samples were only be used for the selectivity and matrix effect test during this method validation and during the analysis of the study samples under Axolabs Project No. AN409.0-24 and will not be used for any other projects.

Table 5: Blank Matrix for selectivity and matrix effect test

Blank Matrix	Sex	Receipt Date	Supplier	Lot No.	Expiry Date	Analysis Batch
Beagle dog CSF	Male	19-Sep-2024	AmplifyBio, West Jefferson, Ohio, USA	102	n.a.	AB01
	Male			103		
	Male			105		
	Female			151		
	Female			152		
	Female			153		

6 PNA Probe

PNA Probe: PNA-BroadInst-01 from HLB Panagene Co., Ltd. (Daejeon, South Korea):

- N-terminus-Atto425 – OO – cdc ttt gtg dgt d – C-terminus, complementary to the antisense strands of 2439-exNA (OO: polyethylene glycol linker; d: 2,6-diaminopurine base modification)
  - Bulk powder was dissolved in 10 % acetonitrile (ACN) and 0.01 % Tween 20
  - Stock solution was diluted with 10 % ACN and 0.01 % Tween 20 to 2 µM (working solution)
- Lot No.: 241164
- Expiry Date: 22-Jul-2027

7 Reference Standard, PNA Probe, and Artificial Matrix – Inventory and Disposition

Records of the receipt, distribution and storage of the reference standard, PNA probe, artificial and biological matrix were maintained.

Unused reference standard will be retained for use in subsequent studies for the Sponsor.

Unused PNA probe and artificial matrix will be retained for use in subsequent studies.

Unused biological matrix was only used for this partial method validation and for the study sample analysis in the corresponding study (Axolabs Project No. AN409.0-24).

Expired reference standard will be disposed of unless the shelf life can be extended (new CoA required).

Expired PNA probe and artificial matrix will be disposed of or transferred to non-GLP storage.

8 METHOD

8.1 Method Summary

This report describes the validation of an AEX-HPLC method with fluorescence detection that allows the sensitive and specific detection of the antisense strands of 2439-exNA, a prion protein (PrP)-lowering divalent siRNA, in dog CSF samples.

The assay is based on the specific hybridization of a 13-mer PNA probe conjugated at the N-terminus with an Atto425 dye that is complementary to the antisense strands of the divalent siRNA 2439-exNA.

The assay is a one-pot approach in which dog CSF lysates are prepared by Proteinase K treatment and the antisense strands of 2439-exNA are hybridized with the PNA in the same step. Subsequently, the samples are analyzed by AEX-HPLC coupled to a fluorescence detector.

Contrary to standard procedure, the addition of 5 % (v/v) of an 8 % Tween 20 and 50 mM TCEP solution directly to the study samples after thawing and before any pipetting step is necessary to prevent nonspecific binding of the analyte to the container surface. Besides that, the working solutions and samples are diluted with 1 % Tween 20 and 1 mM TCEP in artificial CSF. Also, the injected sample volume is 50 µL, resulting in 2.5 µL dog / artificial CSF on column.

Quantification was performed based on an external calibration curve generated from a standard dilution series in artificial CSF. Linear calibration curves (weighted 1/X) were calculated from 1.0 ng/mL to 1,000.0 ng/mL artificial CSF. The concentrations of the calibration standards refer to the concentration of full-length 2439-exNA.

The detailed procedure is described in the STM, which is included as a supplemental file to this report.

8.2 Applicable Documents

- Axolabs Method Validation Plan for the analysis of 2439-exNA in dog CSF (Axolabs Project: AN404.0-24)
- Method Development Report (MDR) *MDR_2439exNA_CSF* (edition: 01, effective date: 21-Aug-2024)
- Standard Test Method (STM) *STM_Broad_2439-exNA_DogCSF* (Revision 01, effective date: 04-Oct-2024)

- KU-RGL-SOP-002, Revision 03 (formerly SOP REG-102) – Methoden-Validierung im GLP/GCP-Bereich (*Method Validation in GLP/GCP Area*)
- KU-RGL-SOP-009, Revision 03 (formerly SOP REG-109) – Durchführung des PNA-HPLC-Assay (*Execution of the PNA-HPLC-Assay*)
  - KU-RGL-INS-005, Revision 05 (formerly REG-109-I01) – PNA-HPLC-Assay SST

8.3 Computerized Systems

Critical computerized systems used in this method validation are listed in Table 6.

Table 6: Computerized Systems

System Name	Revision / Version	System Description
Thermo Fisher Ultimate 3000/Shimadzu LC-2050C HPLC System with a Shimadzu RF-20Axs Fluorescence Detector used with Chromeleon HPLC Software	7.2 SR5	HPLC Data collection and export to Excel
Microsoft Excel Spreadsheet: KU-RGL-ESS-002 (formerly REG-102-S02) – Method Validation Data Report	03	Calculates results of calibration standard and QC sample accuracy and precision
Microsoft Excel Spreadsheet: KU-RGL-ESS-005 (formerly REG-102-S05 – Method Validation Protocol PNA Assay - Liquid Matrices - OnePot	01	Used as instruction and as the protocol for the assay execution for liquid matrices using the OnePot method: Calculates all sample preparation steps and the HPLC sequence for an analysis batch
Microsoft Excel Spreadsheet: KU-RGL-ESS-014 (formerly REG-109-S04 – Preparation of HPLC Buffers & Stock Solutions	02	Calculates all buffer preparation steps, including the preparation of the required stock solutions

8.4 HPLC System and Parameters

The HPLC systems used for the AEX-HPLC analyses are listed in Table 7.

Table 7: HPLC Systems

HPLC System	Module	Supplier	Model	Serial No.
HPLC-GLP-7	Degasser/Solvent Rack	Thermo Fisher Scientific	Ultimate SR-3000	7294808
	Pump		Ultimate LPG-3400SD	8135975
	Autosampler		Ultimate WPS-3000TSL	8135990
	Column Oven		Ultimate TCC-3000RS	6016238
	Fluorescence Detector	Shimadzu	RF-20AXS	L20505573772
	Communication Unit		CBM-20A	L20235570188

- Eluent Buffer A: 25 mM Trizma hydrochloride solution, pH = 8, 1 mM EDTA, 30 % ACN, 3 M urea
- Eluent Buffer B: 25 mM Trizma hydrochloride solution, pH = 8, 1 mM EDTA, 30 % ACN, 3 M urea, 1 M NaBr
- Eluent Buffer C: 10 % ACN, 4 M NaClO4, 2 M urea, 0.5x triphosphate buffer, adjusted to pH 2.5 with phosphoric acid
- Flow Rate: 1 mL/min
- Detection: Excitation @ 436 nm; Emission @ 484 nm; Gain: Medium 16
- Injection of 50 µL of the hybridized sample (corresponds to 2.5 µL matrix on column)
- Column Temperature: 45 °C
- Autosampler Temperature: 20 °C
- HPLC Program: Broad

Table 8: HPLC Gradient Table

Time [min]	% Eluent A	% Eluent B	% Eluent C	Curve
0.00	82	18	0	linear
1.00	82	18	0	linear
5.00	50	50	0	linear
5.10	0	0	100	linear
6.10	0	0	100	linear
6.20	82	18	0	linear
6.70	82	18	0	linear
6.80	0	0	100	linear
7.80	0	0	100	linear
7.90	82	18	0	linear
8.40	82	18	0	linear
8.50	0	0	100	linear
9.50	0	0	100	linear
9.60	82	18	0	linear
13.50	82	18	0	linear

8.5 HPLC Columns

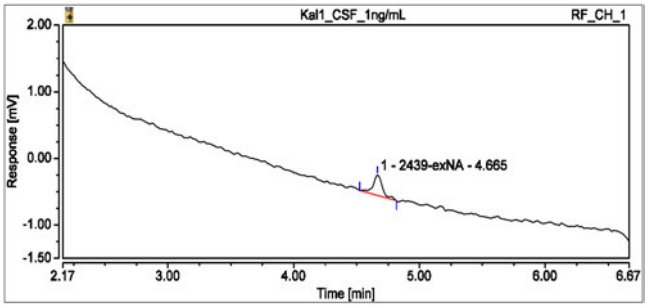
Thermo Fisher Scientific DNAPac PA200 (4 x 250 mm) Column; Serial Number 010628; Lot Number 02306174

8.6 Data Evaluation

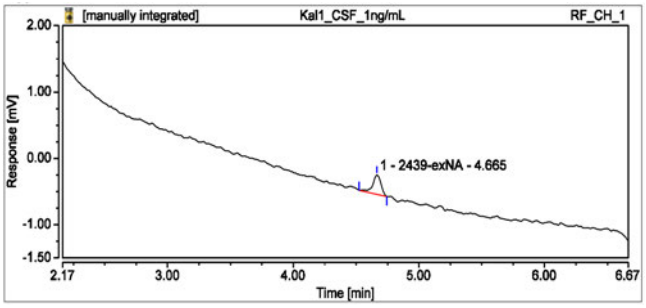
Data acquisition and integration were performed using the software package Chromeleon (Version 7.2) from Thermo Fisher Scientific.

For every injection, the peaks were integrated automatically by the HPLC chromatography software but are adjusted manually if the baseline and peak shoulders are not detected correctly by the HPLC software. When necessary, manual integration was performed by vertical peak splits at minima between two not fully resolved peaks. General rules for manual integration are documented in KU-RGL-SOP-009 (formerly SOP REG-109).

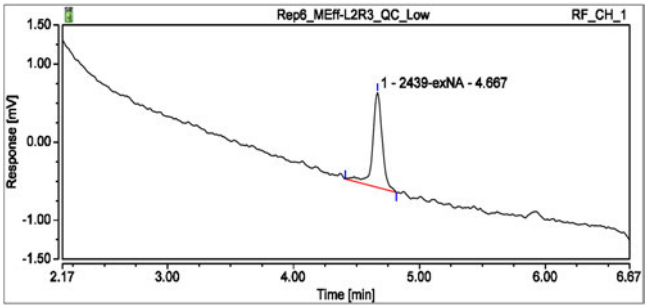
A report format of this file is generated containing values of retention time, peak area, and concentration of 2439-exNA in dog CSF and was saved in PDF format.



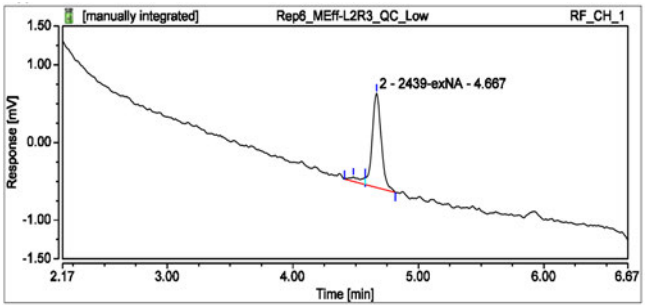
HPLC chromatogram of Cal-level 2 – 1.0 ng/mL, sample no. 5 in AB01 **(original)**



HPLC chromatogram of Cal-level 2 – 1.0 ng/mL, sample no. 5 in AB01 **(re-integrated)**



HPLC chromatogram of Matrix Effect sample QC-Low (3.0 ng/mL), sample no. 32 in AB01 **(original)**



HPLC chromatogram of Matrix Effect sample QC-Low (3.0 ng/mL), sample no. 32 in AB01 **(re-integrated)**

Figure 2: Representative HPLC chromatograms (original and re-integrated)

8.7 Statistical Analysis

Mean values, standard deviations (SD), and coefficients of variation (CV) were calculated using validated Microsoft Excel spreadsheets.

9 Results and Discussion

9.1 System Suitability Test (SST)

The reproducibility of the HPLC system was determined by conducting an SST according to KU-RGL-INS-005 (formerly REG-109-I01) before measuring the analytical batch. Acceptance criteria are described in KU-RGL-INS-005 (formerly REG-109-I01), and the results of the SST run are attached to the documentation of the analysis batch.

A successful SST was completed before the reported analytical batch.

9.2 Calibration Curve

Calibration standards were generated by spiking known concentrations of 2439-exNA into artificial CSF. The linear range was determined from 1.0 ng/mL to 1,000.0 ng/mL. The calibration levels used were 1.0 ng/mL, 5.0 ng/mL, 50.0 ng/mL, 100.0 ng/mL, 500.0 ng/mL, 1,000.0 ng/mL and one blank sample (artificial CSF processed in the absence of 2439-exNA).

The calibration curves were prepared in duplicate for the analytical batch. Blank samples and calibration standards not meeting the acceptance criteria were not used to calculate the calibration curves.

In total 2 calibration curves were analyzed in one analytical batch.

9.2.1 Concentration-Response

The concentration-response relation was calculated by linear regression of the mean peak responses of the 2 calibration standards using 1/X weighting.

The calibration curve parameters (slope, y-axis offset, and coefficient of determination) from reported analysis batches are listed in Table 10.

9.2.2 Calibration Curve Accuracy

Acceptance Criteria:

- The back-calculated concentrations of the calibration standards should be within ± 15 % of the nominal concentrations (± 20 % for the LLOQ calibration standard).
- At least 75 % of the calibration standards, with a minimum of 6 calibration standard levels, and at least 50 % of the calibration standards per concentration level must fulfill this criterion.

Results:

- 100.0 % of the back-calculated concentrations for all calibration standards were within ± 15 % (± 20 % for the LLOQ calibration standard) of the nominal concentrations.

Data are listed in Table 11.

Based on these results, the method can be used to determine concentrations of 2439-exNA in artificial / dog CSF over the concentration range of 1.0 ng/mL to 1,000.0 ng/mL.

9.2.3 Sensitivity/Lower Limit of Quantification (LLOQ)

The sensitivity of the assay is defined as the lowest 2439-exNA concentration that can be quantified with acceptable accuracy and precision – i.e., the LLOQ.

To confirm the LLOQ, 2 individually prepared samples with a nominal concentration at the LLOQ were analyzed (LLOQ calibration standards).

Acceptance Criteria:

- The analyte response (peak area) in each LLOQ calibration standard should be at least 5 times higher than the response (peak area) at the retention time of the analyte in the blank sample of the same calibration curve.
- The mean back-calculated concentration of the 2 LLOQ calibration standards should be within ± 20 % of the nominal concentration and the coefficient of variation (CV) should not exceed 20 %.

Results:

- No signal at the retention time of 2439-exNA was detected in the blank samples of the calibration curves and hence, the 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the blank sample of the same calibration curve.
- The mean back-calculated concentration of the 2 LLOQ calibration standards was 105.0 % of the nominal concentration with a CV of 20.2 %.

Data are listed in Table 11 and Table 12.

The LLOQ for 2439-exNA was confirmed at 1.0 ng/mL in artificial CSF. The LLOQ can also be applied when using dog CSF.

9.2.4 Upper Limit of Quantification (ULOQ)

The highest calibration standard in the calibration curve defines the ULOQ.

Two (2) individually prepared samples with a nominal concentration at the ULOQ were analyzed (ULOQ calibration standards).

Acceptance Criteria:

The mean back-calculated concentration of the 2 ULOQ calibration standards should be within ± 15 % of the nominal concentration and the CV should not exceed 15 %.

Results:

The mean back-calculated concentration of the 2 ULOQ calibration standards was 100.0 % of the nominal concentration with a CV of 0.4 %.

Data are listed in Table 11.

The ULOQ for 2439-exNA was confirmed at 1,000.0 ng/mL in artificial CSF. The ULOQ can also be applied when using dog CSF.

9.3 Selectivity

The selectivity of the method was evaluated by comparison of unspiked samples of 6 different blank dog CSF lots with the same dog CSF samples spiked with 2439-exNA at the concentration of the LLOQ (QC-LLOQ).

Acceptance Criteria:

- The analyte response (peak area) in each QC-LLOQ sample should be at least 5 times higher than the response (peak area) at the retention time of the analyte in the blank sample of the same matrix lot.
- The back-calculated concentration each QC-LLOQ sample should be within ± 20 % of the nominal concentration.

Results:

- No signal at the retention time of 2439-exNA was detected in the selectivity blank samples and hence, the 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the blank sample of the same matrix lot.
- The back-calculated concentrations of the QC-LLOQ samples ranged from 110.0 % – 120.0 % of the nominal concentration.

Data are listed in Table 13.

Selectivity of the method was confirmed for dog CSF.

9.4 Matrix Effect

The matrix effect was evaluated by analyzing triplicates of QC-Low and QC-High samples, each prepared using 6 different blank dog CSF lots.

Acceptance Criteria:

For each evaluated blank dog CSF lot, the mean back-calculated concentration of the QC-Low and QC-High samples (each triplicates) should be within ± 15 % of the nominal concentrations and the CV should not exceed 15 %.

Results:

- For 6 matrix lots (102, 103, 105, 151, 152, 153), the mean back-calculated concentrations of the QC samples ranged from 102.2 % – 107.8 % (QC-Low) and 98.8 % – 100.8 % (QC-High) of the nominal concentrations, and CV values ranged from 3.1 % – 4.9 % (QC-Low) and 0.1 % – 1.3 % (QC-High).

Data are listed in Table 14.

Based on these results, there is no matrix effect for the assay regarding the analysis of 2439-exNA in dog CSF.

9.5 Carry-Over

Blank samples were injected after the ULOQ calibration standards to assess carry-over.

Acceptance Criteria:

The analyte response (peak area) in each LLOQ calibration standard should be at least 5 times higher than the analyte response (peak area) in the carry-over blank sample following the ULOQ calibration standard of the same calibration curve.

Results:

- No signal at the retention time of 2439-exNA was detected in the carry-over blank sample and hence, the 2439-exNA response (peak area) in each LLOQ calibration standard was at least 5 times higher than the response (peak area) in the carry-over blank sample following the ULOQ calibration standard of the same calibration curve.

Data are listed in Table 15.

No carry-over was observed for 2439-exNA in artificial CSF. The result can also be applied when using dog CSF. Therefore, no special precautions after the analysis of samples with a high concentration of 2439-exNA have to be implemented.

9.6 Quality Control (QC) Samples

Analysis of assay accuracy and precision, dilution integrity, and stability was conducted by analysis of QC samples of defined concentrations.

QC samples were prepared by spiking known amounts of 2439-exNA into artificial CSF.

QC samples were prepared at 4 concentration levels which cover the calibration curve range:

- at the LLOQ (QC-LLOQ): 1.0 ng/mL
- within 3 times above the LLOQ (QC-Low): 3.0 ng/mL
- at 30 – 50 % of the calibration curve range (QC-Mid): 450.0 ng/mL
- at least at 75 % of the upper calibration curve range (QC-High): 900.0 ng/mL

9.7 Accuracy

The accuracy of the analytical method describes the closeness of the determined concentration obtained by the method to the nominal concentration of 2439-exNA (expressed in percentage).

The analysis of assay accuracy was conducted using the QC samples at the 4 concentration levels that were measured as 6 individually prepared replicates in the analytical run.

The concentration of 2439-exNA was calculated against the external calibration curve, generated in duplicate.

9.7.1 Within-Run Accuracy

Acceptance Criteria:

The mean back-calculated concentration of the QC samples at each of the 4 concentration levels should be within ± 15 % of the nominal concentrations (± 20 % for the QC-LLOQ).

Results:

The mean back-calculated concentrations of the QC samples at each of the 4 concentration levels is 111.7 % (QC-LLOQ), 102.2 % (QC-Low), 99.7 % (QC-Mid) and 97.1 % (QC-High) of the nominal concentrations.

Data are listed in Table 16.

Based on these results, the assay for 2439-exNA in artificial CSF is considered accurate. The result can also be applied when using dog CSF.

9.7.2 Between-Run Accuracy

The between-run accuracy was assessed in Method Validation AN405.0-24 conducted for 2439-exNA in rat CSF, using artificial CSF.

9.8 Precision

The precision of the analytical method describes the closeness of repeated individual measures of the 2439-exNA concentration, expressed as coefficient of variation (CV).

The analysis of assay precision was conducted using the QC samples at the 4 concentration levels. All concentration levels were measured as 6 individually prepared replicates in the analytical run.

The same analytical run and data were used for the demonstration of accuracy.

The concentration of 2439-exNA was calculated against the external calibration curve, generated in duplicate.

9.8.1 Within-Run Precision

Acceptance Criteria:

The within-run CV of the QC samples at each of the 4 concentrations levels should not exceed 15 % (20 % for the QC-LLOQ).

Results:

The within-run CV of the QC samples at each of the 4 concentration levels is 6.7 % (QC-LLOQ), 2.7 % (QC-Low), 3.0 % (QC-Mid) and 5.9 % (QC-High).

Data are listed in Table 16.

Based on these results, the assay for 2439-exNA in artificial CSF is considered precise. The result can also be applied when using dog CSF.

9.8.2 Between-Run Precision

The between-run precision was assessed in Method Validation AN405.0-24 conducted for 2439-exNA in rat CSF, using artificial CSF.

9.9 Dilution Integrity

Dilution integrity was assessed in Method Validation AN405.0-24 conducted for 2439-exNA in rat CSF, using artificial CSF.

9.10 Stability

The stability of 2439-exNA was analyzed in Method Validation AN405.0-24 conducted for 2439-exNA in rat CSF. Since both projects use artificial CSF as a surrogate matrix, the tests would have been identical and therefore superfluous.

10 Deviations

The bioanalytical method validation for the analysis of 2439-exNA in dog CSF was conducted according to the Method Validation Plan and to applicable SOPs without any deviation.

11 Conclusions

The bioanalytical method for the analysis of 2439-exNA in dog CSF was successfully validated and is described in the *STM_Broad_2439-exNA_DogCSF* (Revision 01, effective date: 04-Oct-2024). All validation tests fulfilled the acceptance criteria described in the Method Validation Plan.

The method can be used within a linear range from 1.0 ng/mL to 1,000.0 ng/mL of 2439-exNA in artificial or dog CSF, applying a 1/X weighted linear regression model.

The stability of 2439-exNA was analyzed in Method Validation AN405.0-24 conducted for 2439-exNA in rat CSF. Since both projects use artificial CSF as a surrogate matrix, the tests would have been identical. Therefore, stability results will also be reported in the Method Validation AN405.0-24.

12 List of Records to be Retained

All original data and documents generated during this method validation will be archived at Axolabs GmbH or at an Axolabs-employed GLP-compliant archive facility under appropriate conditions in accordance with German and EU legislation, and KU-RGL-SOP-005 (formerly REG-105).

The following records will be retained:

- Final signed Partial Method Validation Plan
- Critical communication, as appropriate
- Raw data:
  - HPLC files with HPLC chromatograms, audit trails, and HPLC program
  - Assay protocols
  - System Suitability Test reports
  - CoA of PNA probe and reference standard
  - Other raw data (if applicable)
- QC and QA records of document and data reviews/audits
- Final signed Method Validation Report and all addenda

A backup file of all analytical HPLC data, together with an electronic PDF report of all analytical runs, and all electronic files generated during the method validation (reports, Excel-Files) are stored in electronic form.

13 Data Tables

Table 9: Analytical Batches for the Analysis of 2439-exNA in Dog CSF

Run ID	Date of Analysis	Description	Status	Comments
AN404.0-24_01_AB01	20-Sep-2024	Validationbatch with calibration curve and accuracy/precision in artificial CSF, matrix effect and selectivity in dog CSF	passed	n.a.

Table 10: Calibration Curve Parameters for the Analysis of 2439-exNA in Artificial CSF

Equation for the calibration curve using a linear regression model weighted 1/X: y = m*x + b					
Run ID	**Date of Analysis**	**Sample No.**	**Slope [m]**	**Y-Offset [b]**	**R-Squared**
AB01	20-Sep-2024	5-10, 94-99	0.03420	-0.00956	0.99974

Table 11: Results of the Calibration Standards for the Analysis of 2439-exNA in Artificial CSF

		Cal-Level-2 (LLOQ)		Cal-Level-3		Cal-Level-4		Cal-Level-5		Cal-Level-6		Cal-Level-7 (ULOQ)	
		1.0 ng/mL		5.0 ng/mL		50.0 ng/mL		100.0 ng/mL		500.0 ng/mL		1,000.0 ng/mL	
Run ID	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB01	5 – 10	0.9	90.0 %	5.0	100.0 %	47.3	94.6 %	91.2	91.2 %	493.5	98.7 %	997.0	99.7 %
AB01	94 – 99	1.2	120.0 %	4.6	92.0 %	50.9	101.8 %	103.3	103.3 %	514.6	102.9 %	1,002.5	100.3 %
	Mean	1.05	105.0 %	4.80	96.0 %	49.10	98.2 %	97.25	97.3 %	504.05	100.8 %	999.75	100.0 %
	SD (±)	0.21		0.28		2.55		8.56		14.92		3.89	
	CV (%)	20.2 %		5.9 %		5.2 %		8.8 %		3.0 %		0.4 %	
	N	2		2		2		2		2		2	

Table 12: Sensitivity/Lower Limit of Quantification (LLOQ) for the Analysis of 2439-exNA in Artificial CSF

		Cal-Level-1 (Blank Sample)		Cal-Level-2 (LLOQ)	
		0.0 ng/mL		1.0 ng/mL	
Run ID	**Sample No.**	**Peak Area [mV*min]**	**Acc. Criterion ≤ 20 % of LLOQ**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**
AB01	4, 5	0.0000	0.0 %	0.0223	0.9
AB01	93, 94	0.0000	0.0 %	0.0323	1.2

Table 13: Selectivity for the Analysis of 2439-exNA in Dog CSF

					Blank Sample for Selectivity		QC-LLOQ		
					0.0 ng/mL		1.0 ng/mL		
Run ID	**Date of Analysis**	**Sex**	**Blank Matrix Lot No.**	**Sample No.**	**Peak Area [mV*min]**	**Acc. Criterion ≤ 20 % of LLOQ**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**
AB01	20-Sep-2024	Male	102	13, 20	0.0000	0.0 %	0.0304	1.2	120.0 %
			103	14, 21	0.0000	0.0 %	0.0299	1.2	120.0 %
			105	15, 22	0.0000	0.0 %	0.0310	1.2	120.0 %
		Female	151	16, 23	0.0000	0.0 %	0.0278	1.1	110.0 %
			152	17, 24	0.0000	0.0 %	0.0310	1.2	120.0 %
			153	18, 25	0.0000	0.0 %	0.0302	1.2	120.0 %

Table 14: Matrix Effect for the Analysis of 2439-exNA in Dog CSF

			QC-Low		QC-High	
			3.0 ng/mL		900.0 ng/mL	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB01	102	27, 46	3.2	106.7 %	909.6	101.1 %
		28, 47	3.1	103.3 %	901.4	100.2 %
		29, 48	3.3	110.0 %	909.6	101.1 %
		Mean	3.20	106.7 %	906.87	100.8 %
		SD (±)	0.10		4.73	
		CV (%)	3.1 %		0.5 %	
		N	3		3	
AB01	103	30, 49	3.1	103.3 %	897.0	99.7 %
		31, 50	3.3	110.0 %	902.3	100.3 %
		32, 51	3.2	106.7 %	898.9	99.9 %
		Mean	3.20	106.7 %	899.40	99.9 %
		SD (±)	0.10		2.69	
		CV (%)	3.1 %		0.3 %	
		N	3		3	

Table 14: Matrix Effect for the Analysis of 2439-exNA in Dog CSF (Continued)

			QC-Low		QC-High	
			3.0 ng/mL		900.0 ng/mL	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB01	105	33, 52	3.3	110.0 %	901.1	100.1 %
		34, 53	3.1	103.3 %	892.7	99.2 %
		35, 54	3.3	110.0 %	893.4	99.3 %
		Mean	3.23	107.8 %	895.73	99.5 %
		SD (±)	0.12		4.66	
		CV (%)	3.6 %		0.5 %	
		N	3		3	
AB01	151	36, 55	3.3	110.0 %	877.4	97.5 %
		37, 56	3.1	103.3 %	890.2	98.9 %
		38, 57	3.0	100.0 %	899.9	100.0 %
		Mean	3.13	104.4 %	889.17	98.8 %
		SD (±)	0.15		11.29	
		CV (%)	4.9 %		1.3 %	
		N	3		3	

Table 14: Matrix Effect for the Analysis of 2439-exNA in Dog CSF (Continued)

			QC-Low		QC-High	
			3.0 ng/mL		900.0 ng/mL	
Run ID	**Blank Matrix Lot No.**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**
AB01	152	39, 58	3.1	103.3 %	890.4	98.9 %
		40, 59	3.2	106.7 %	892.3	99.1 %
		41, 60	3.0	100.0 %	888.1	98.7 %
		Mean	3.10	103.3 %	890.27	98.9 %
		SD (±)	0.10		2.10	
		CV (%)	3.2 %		0.2 %	
		N	3		3	
AB01	153	42, 61	3.0	100.0 %	892.9	99.2 %
		43, 62	3.0	100.0 %	891.1	99.0 %
		44, 63	3.2	106.7 %	893.3	99.3 %
		Mean	3.07	102.2 %	892.43	99.2 %
		SD (±)	0.12		1.17	
		CV (%)	3.8 %		0.1 %	
		N	3		3	

Table 15: Carry-Over for the Analysis of 2439-exNA in Artificial CSF

		Cal-Level-2 (LLOQ)			Cal-Level-7 (ULOQ)			Blank Sample for Carry-Over		
		1.0 ng/mL			1,000.0 ng/mL			0.0 ng/mL		
Run ID	**Sample No.**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**	**Peak Area [mv*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Peak Area [mV*min]**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ≤ 20 % of LLOQ**
AB01	5, 10, 11	0.0223	0.9	90.0 %	34.0856	997.0	99.7 %	0.0000	BQL	0.0 %
AB01	94, 99, 100	0.0323	1.2	120.0 %	34.2737	1,002.5	100.3 %	0.0000	BQL	0.0 %

Table 16: Results of the QC Samples (Accuracy and Precision) for the Analysis of 2439-exNA in Artificial CSF

					QC-LLOQ		QC-Low		QC-Mid		QC-High	
					1.0 ng/mL		3.0 ng/mL		450.0 ng/mL		900.0 ng/mL	
Run ID	**Date of Analysis**	**User**	**HPLC System**	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 20 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ± 15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion ±15 % of Nominal**
AB01	20-Sep-2024	JSt	HPLC-GLP-7	65, 72, 79, 86	1.2	120.0 %	3.1	103.3 %	450.8	100.2 %	909.2	101.0 %
				66, 73, 80, 87	1.0	100.0 %	3.1	103.3 %	452.0	100.4 %	891.0	99.0 %
				67, 74, 81, 88	1.1	110.0 %	3.1	103.3 %	441.4	98.1 %	892.9	99.2 %
				68, 75, 82, 89	1.1	110.0 %	3.1	103.3 %	425.9	94.6 %	910.6	101.2 %
				69, 76, 83, 90	1.1	110.0 %	3.1	103.3 %	459.6	102.1 %	775.2	86.1 %
				70, 77, 84, 91	1.2	120.0 %	2.9	96.7 %	462.0	102.7 %	863.4	95.9 %
Accuracy/Precision				**Mean**	1.12	111.7 %	3.07	102.2 %	448.62	99.7 %	873.72	97.1 %
				SD (±)	0.08		0.08		13.30		51.18	
				CV (%)	6.7 %		2.7 %		3.0 %		5.9 %	
				N	6		6		6		6	

14 SUPPLEMENTARY INFORMATION

14.1 Supplemental File 1

Axolabs *MDR_2439exNA_CSF*

14.2 Supplemental File 2

Axolabs *STM_Broad_2439-exNA_DogCSF*